

BREAST CANCER SEGMENTATION USING HYBRID U-NET AND NSCGCN ARCHITECTURE

**A REPORT SUBMITTED IN
PARTIAL FULFILMENT OF THE REQUIREMENTS FOR
INTERNSHIP**

**By
Snigdha Gorai
VIT-AP University**

**DURING
1st May to 30th June**

**UNDER THE GUIDANCE OF
Kanchan Lata Kashyap**

**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
KOTTAYAM**



CERTIFICATE

This is to certify that the work contained in this internship report entitled "BREAST CANCER SEGMENTATION USING HYBRID U-NET AND NSCGCN ARCHITECTURE" submitted by Snigdha Gorai to the Indian Institute of Information Technology Kottayam as part of the Summer Internship Program has been carried out by her under my supervision and that it has not been submitted elsewhere for the award of any degree or diploma.

The assistance and help during the course of this internship from colleagues and other sources has been properly acknowledged. The contents and results presented in this report are based on the candidate's own work.

Kottayam – 686635
June 2026

Mentor
Kanchan Lata Kashyap
Supervisory Guide

DECLARATION

I, Snigdha Gorai, hereby declare that the report entitled "BREAST CANCER SEGMENTATION USING HYBRID U-NET AND NSCGCN ARCHITECTURE" submitted to the Indian Institute of Information Technology Kottayam as part of the Summer Internship Program is an original work carried out by me under the supervision of Kanchan Lata Kashyap. I also certify that all information, data, and references used in this report have been acknowledged properly. I confirm that the content of this report is a true representation of the work completed during my internship.

I also declare that this report has not been submitted, either wholly or in part, to any other university or institution for any other degree or diploma.

The research work presented herein is based on honest observations and findings obtained through systematic experimentation and analysis of data during the internship period.



Snigdha Gorai
VIT-AP University

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my internship mentor, Kanchan Lata Kashyap, for her invaluable guidance, constant encouragement, and constructive feedback throughout the course of this internship. Her expertise in the field of deep learning and medical image analysis was instrumental in shaping this project.

I am also thankful to Indian Institute of Information Technology Kottayam for providing the computational resources and infrastructure necessary for conducting the experiments in this study. The access to high-performance GPU servers was essential for training the deep learning models in a timely manner.

I extend my thanks to my peers and colleagues at the institute who provided insightful discussions and helped me overcome various technical challenges encountered during the course of this work.

Finally, I am grateful to my family for their unwavering support and encouragement during the entire internship period.

Snigdha Gorai

ABSTRACT

Breast cancer detection and segmentation is an important aspect of medical image analysis, requiring efficient models to accurately identify malignant regions in ultrasound images. This report presents a study of a deep learning pipeline using a hybrid architecture: a standard U-Net combined with a Non-Local Spatial Context Graph Convolutional Network (NSCGCN). The primary objective is to evaluate the performance of this hybrid model, particularly focusing on the number of nodes in the graph projection, to accurately segment breast cancer lesions.

While standard Convolutional Neural Networks (CNNs), like U-Net, have shown great success in medical image segmentation, they are fundamentally limited by their local receptive fields. They often struggle to capture long-range spatial dependencies and global context, which can be critical for accurately distinguishing lesion boundaries from normal tissue in complex ultrasound images. To overcome this limitation, this project employs a hybrid architecture integrating the NSCGCN at the U-Net bottleneck for global reasoning over feature maps.

In this study, an ablation analysis was performed on the NSCGCN module. The configuration utilizing 32 nodes achieved the highest performance with a Dice score of 77.61% and an Intersection over Union (IoU) of 67.30%, showcasing its effectiveness in capturing long-range spatial dependencies compared to other node configurations (16 and 128 nodes).

Keywords: Breast Cancer Segmentation, U-Net, Graph Convolutional Network, NSCGCN, Medical Image Analysis, Deep Learning, Ablation Study, Computer-Aided Diagnosis.

TABLE OF CONTENTS

Certificate

Declaration

Acknowledgement

Abstract

List of Figures

1. Introduction

1.1 The Problem

1.2 Our Approach: U-Net + NSCGCN

1.3 Objectives

2. Literature Review

2.1 U-Net for Biomedical Image Segmentation

2.2 Non-Local Neural Networks

2.3 Graph Convolutional Networks in Segmentation

3. Methodology

3.1 Datasets

3.2 Preprocessing

3.3 Model Architecture

3.4 Model Training

3.5 Model Evaluation

4. Result and Discussion

4.1 Ablation Study Results

4.2 Confusion Matrix Analysis

4.3 Discussion

5. Conclusion and Impact

6. References

LIST OF FIGURES

- Figure 1:** Sample breast ultrasound image showing a lesion
- Figure 2:** Segmentation example — Input, Ground Truth, and Prediction
- Figure 3:** Data augmentation techniques applied to ultrasound images
- Figure 4:** End-to-end workflow of the proposed hybrid model pipeline
- Figure 5:** Proposed UNetGCN encoder-decoder architecture with NSCGCN bottleneck
- Figure 6:** Original U-Net architecture with skip connections
- Figure 7:** NSCGCN module — Projection, Graph Reasoning, and Reprojection
- Figure 8:** Graph Convolutional Network concept
- Figure 9:** Training and validation learning curves over 50 epochs
- Figure 10:** Ablation study results — Performance metrics comparison (bar chart)
- Figure 11:** Ablation study — Line chart showing effect of node count
- Figure 12:** Radar chart — Multi-metric comparison across configurations
- Figure 13:** Confusion Matrix — 16 Nodes configuration
- Figure 14:** Confusion Matrix — 32 Nodes configuration
- Figure 15:** Confusion Matrix — 128 Nodes configuration

1. INTRODUCTION

1.1 The Problem

Breast cancer is one of the most prevalent cancers worldwide and a leading cause of cancer-related deaths among women. According to the World Health Organization (WHO), breast cancer accounts for approximately 12.5% of all new annual cancer cases globally. Early detection and accurate diagnosis are paramount for successful treatment and improved survival rates, making efficient screening methods an area of critical research.

Ultrasound imaging is a widely used modality for breast examination due to its non-invasive nature, cost-effectiveness, and real-time capabilities. Unlike mammography, it does not expose patients to ionizing radiation, making it particularly suitable for younger women and for use as a supplementary screening tool. However, reading ultrasound images is highly dependent on the operator's experience, and the images often suffer from speckle noise, low contrast, and ambiguous boundaries. Automated, accurate segmentation of lesions in ultrasound images remains a challenging yet crucial task for computer-aided diagnosis (CAD) systems that can significantly aid radiologists in their clinical workflows.

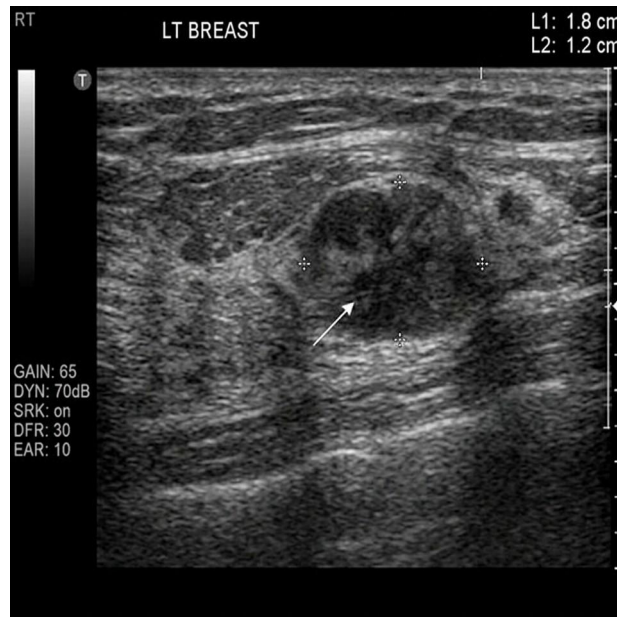


Figure 1: Sample breast ultrasound image showing a lesion

1.2 Our Approach: U-Net + NSCGCN

While standard Convolutional Neural Networks (CNNs), like the well-known U-Net, have shown great success in medical image segmentation, they are fundamentally limited by their local receptive fields. They often struggle to capture long-range spatial dependencies and global context, which can be critical for accurately distinguishing lesion boundaries from normal tissue in complex ultrasound images.

To overcome this limitation, this project employs a hybrid architecture: U-Net integrated with a Non-Local Spatial Context Graph Convolutional Network (NSCGCN). The approach works in three key stages:

- **Feature Extraction:** The standard U-Net encoder extracts hierarchical local feature maps from the input ultrasound images, capturing low-level details like edges and textures as well as higher-level semantic features.
- **Global Reasoning:** The NSCGCNModule is strategically placed at the bottleneck of the U-Net. Instead of just looking at local patches, the NSCGCN constructs a graph from the feature maps, mapping spatial features into an interaction space. This allows the network to perform global reasoning, capturing long-range dependencies and contextual relationships across the entire image.
- **Refined Segmentation:** The decoder part of the U-Net then reconstructs the segmentation mask, utilizing both the local features from skip connections and the globally-aware features from the NSCGCN module. This combination results in more accurate and robust segmentation boundaries compared to using purely local convolutional filters.

Breast Ultrasound Image Segmentation Example

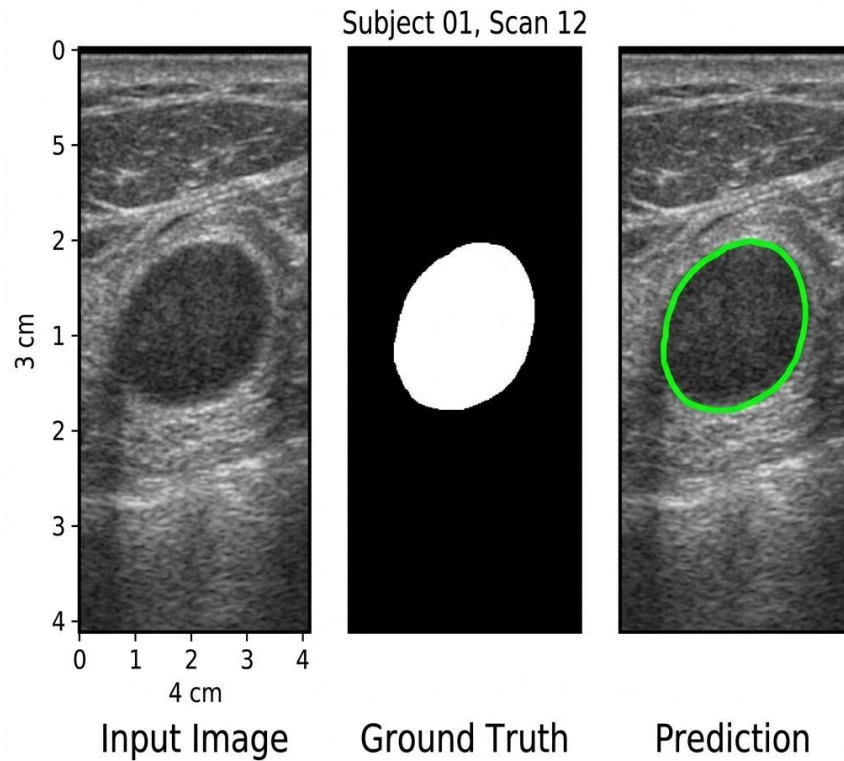


Figure 2: Segmentation example — Input Image, Ground Truth Mask, and Model Prediction

1.3 Objectives

The primary objectives of this study are:

- **(i)** To implement and evaluate a hybrid U-Net + NSCGCN architecture for breast cancer segmentation from ultrasound images.
- **(ii)** To conduct an ablation study on the number of graph nodes (16, 32, 128) in the NSCGCN module to determine the optimal configuration.
- **(iii)** To analyze the impact of global graph-based reasoning on segmentation accuracy compared to purely local convolutional approaches.
- **(iv)** To contribute toward more reliable and efficient computer-aided diagnosis tools that can assist healthcare professionals in making better-informed, timely decisions for patient care.

2. LITERATURE REVIEW

During the internship, several foundational works on medical image segmentation and graph-based reasoning were reviewed. The key papers that influenced the design of this study are discussed below.

2.1 U-Net for Biomedical Image Segmentation

One notable study, "U-Net: Convolutional Networks for Biomedical Image Segmentation" by Olaf Ronneberger et al. (2015), proposed the U-Net architecture, which has become the de facto standard for medical image segmentation tasks. The architecture features a symmetric encoder-decoder structure with skip connections that concatenate feature maps from the encoder to the decoder at corresponding resolution levels. This design allows the network to recover fine-grained spatial details that are typically lost during downsampling operations. The U-Net was originally designed for electron microscopy images but has since been applied successfully to a wide variety of medical imaging modalities including CT, MRI, and ultrasound.

2.2 Non-Local Neural Networks

Another significant contribution in the field of non-local reasoning is "Non-local Neural Networks" by Xiaolong Wang et al. (2018). This work highlighted the importance of capturing long-range dependencies in images—something standard convolutions struggle with due to their localized receptive fields. The non-local operation computes the response at a position as a weighted sum of features at all positions in the feature map. This concept is analogous to the self-attention mechanism in transformers and has been shown to improve performance in tasks requiring global context understanding.

2.3 Graph Convolutional Networks in Segmentation

Recent advancements in combining GCNs with CNNs for semantic segmentation have shown that projecting spatial features into a graph domain allows for efficient global context aggregation. Traditional GCN approaches, however, suffer from the well-known over-smoothing problem, where node representations become indistinguishable as the number of layers increases. Our study builds upon these concepts by implementing a Node-Self Convolution (NSC), which

explicitly maintains node identity through a dedicated self-weight matrix, effectively mitigating the over-smoothing issue while retaining the benefits of global reasoning.

3. METHODOLOGY

3.1 Datasets

We used a breast ultrasound dataset split into three distinct directories: train, valid, and test. The dataset comprises original ultrasound images and their corresponding ground truth binary masks. Each mask contains pixel-level annotations where the lesion region is marked as foreground (white) and the surrounding tissue as background (black). The images are stored in a standard image format and are processed to extract the regions of interest accurately.

The dataset split distribution is shown in the table below:

Split	No. of Images	Purpose
Train	~70%	Model parameter learning
Validation	~15%	Hyperparameter tuning & early stopping
Test	~15%	Final performance evaluation

3.2 Preprocessing

All images and masks in the datasets were resized into a uniform resolution of 256×256 pixels to match the input requirements of the models. For the training process, data augmentation techniques using the Albumentations library were applied to improve the diversity of the training data and reduce overfitting. These augmentations included:

- **Flips:** Horizontal and Vertical Flips (with 50% probability each).
- **Geometric:** Shift, Scale, and Rotate transformations (shift limit: 0.1, scale limit: 0.1, rotate limit: 15°).
- **Normalization:** Image normalization using standard ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

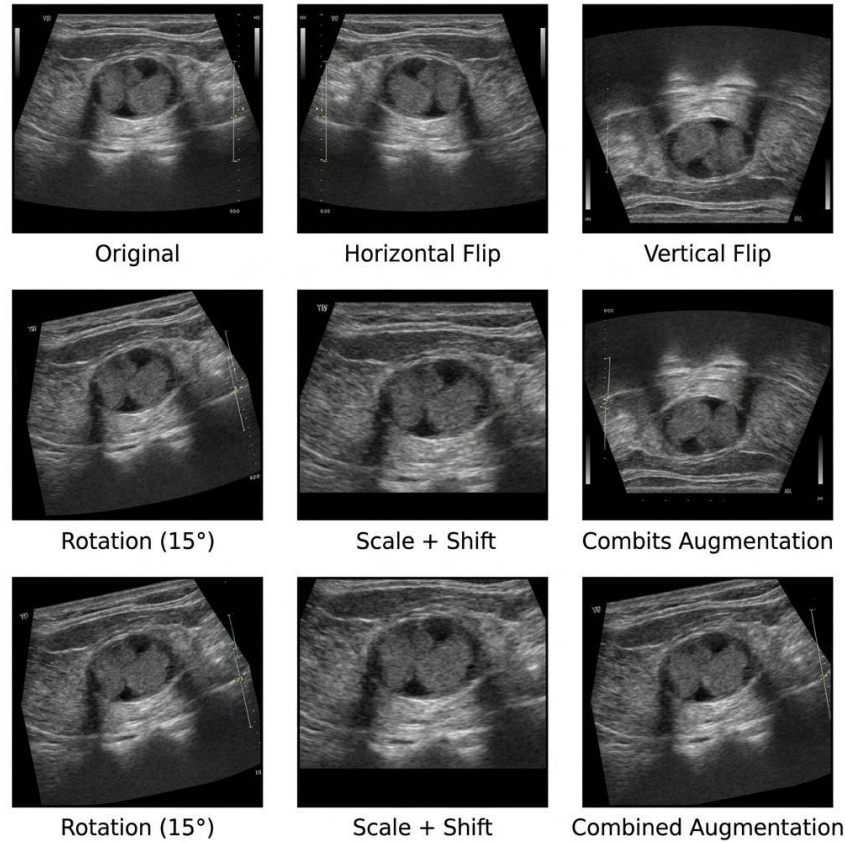


Figure 3: Data augmentation techniques applied to breast ultrasound images

Data augmentation is a critical step in medical image segmentation, as medical datasets are typically small. By artificially increasing the diversity of the training data, augmentation helps the model generalize better to unseen images and reduces the risk of overfitting to the limited training samples.

3.3 Model Architecture

We focused on a hybrid architecture combining a U-Net and a custom GCN module. The complete workflow of the proposed pipeline is illustrated below:

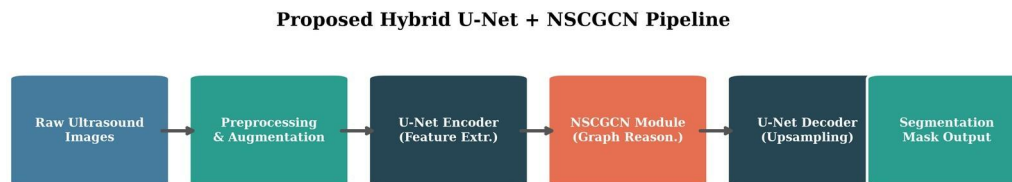


Figure 4: End-to-end workflow of the proposed hybrid model pipeline

The model (UNetGCN) uses a standard U-Net encoder-decoder structure with skip connections. To capture long-range spatial dependencies that are missed by the limited receptive field of convolutional layers, an NSCGCNModule is strategically placed at the bottleneck of the U-Net. Instead of just looking at local patches, the NSCGCN constructs a graph from the feature maps, mapping spatial features into an interaction space. This allows the network to perform global reasoning, capturing long-range dependencies and contextual relationships across the entire image.

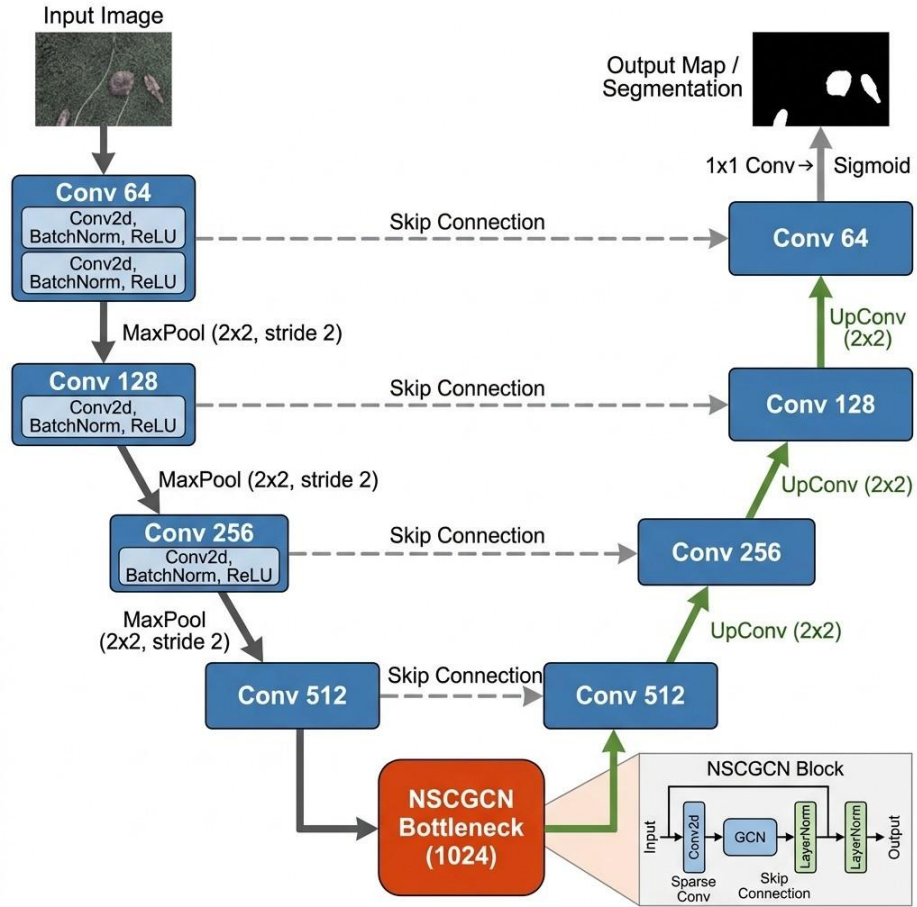


Figure 5: Proposed UNetGCN architecture — encoder-decoder with NSCGCN bottleneck

The original U-Net architecture that forms the backbone of our model:

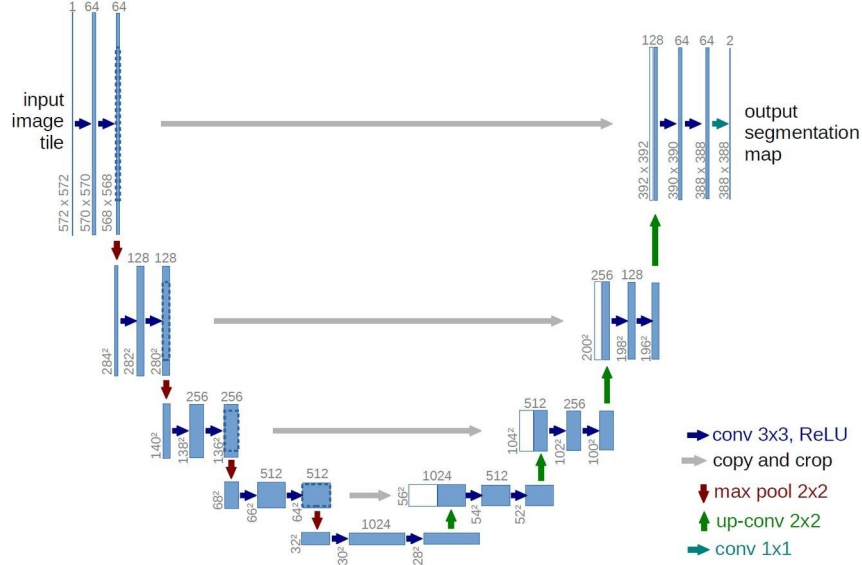


Figure 6: Original U-Net architecture with encoder-decoder structure and skip connections

The NSCGCN module operates in three stages:

- **Projection:** Spatial features ($H \times W \times C$) are projected into a fixed number of graph nodes (N) using a learned projection matrix B . This maps each spatial location to a soft assignment over graph nodes, transforming the spatial domain into a graph domain.
- **Node-Self Convolution (NSC):** Unlike standard GCNs which can over-smooth features, the NSC layer explicitly maintains node identity by fusing standard neighbor message passing (using weight W_{neighbor}) with a dedicated node-self convolution weight (W_{self}). This preserves distinctive node features while still enabling global information exchange across all nodes.
- **Reprojection:** The refined graph node features are projected back into the spatial domain using the transpose of the projection matrix (B^T), producing enhanced feature maps that are globally aware and are passed to the U-Net decoder for reconstruction.

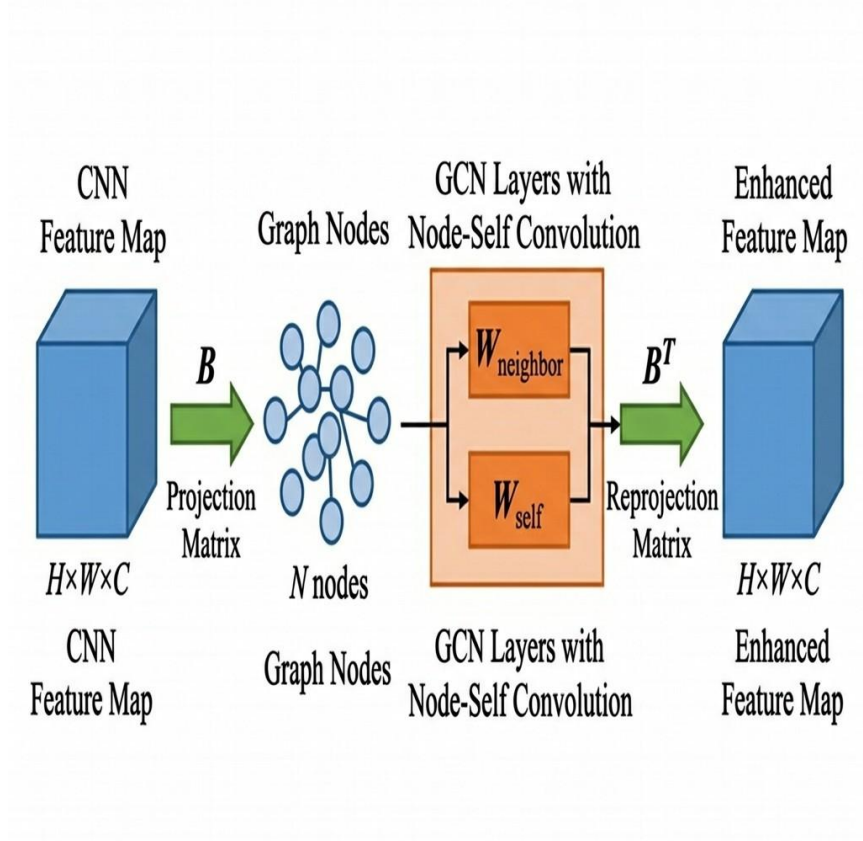


Figure 7: NSCGCN module — Projection into graph domain, Graph Reasoning, and Reprojection

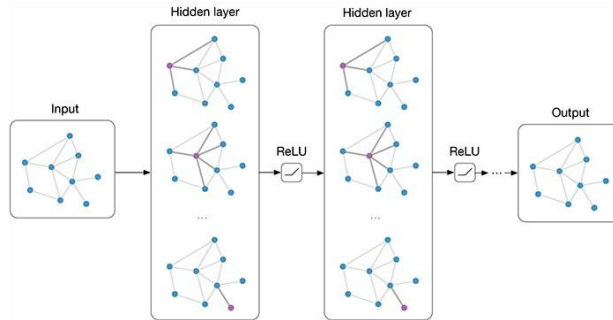


Figure 8: Graph Convolutional Network concept — nodes aggregate information from neighbors

3.4 Model Training

The model was compiled using the AdamW optimizer with a learning rate of $1e-3$ and a weight decay of $1e-4$. The loss function used was a combination of Binary Cross Entropy (BCE) and Dice Loss (BCEDiceLoss), which is highly effective for addressing the class imbalance inherent in segmentation tasks where background pixels vastly outnumber foreground pixels.

The models were trained for 50 epochs with a batch size of 8. Training was monitored using a validation split, and a ReduceLROnPlateau scheduler was used to halve the learning rate if the validation Dice score plateaued for 5 consecutive epochs. The best model weights were saved based on the highest validation Dice score achieved during training (checkpoint mechanism).

The complete training hyperparameters are summarized in the table below:

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	1e-3
Weight Decay	1e-4
Loss Function	BCE + Dice Loss
Epochs	50
Batch Size	8
LR Scheduler	ReduceLROnPlateau (patience=5, factor=0.5)
Image Size	256 × 256 pixels

The training and validation learning curves over 50 epochs are shown below. The loss curve (left) shows a steady decrease in both training and validation loss, while the Dice score curve (right) shows the corresponding improvement in segmentation quality:

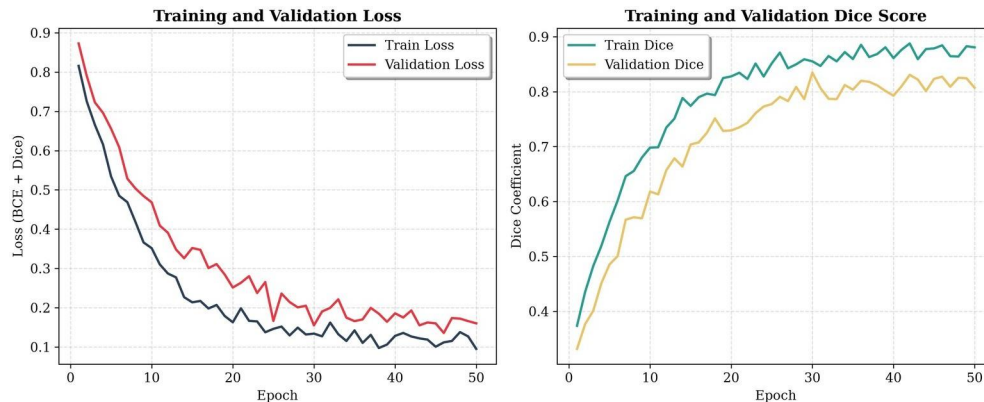


Figure 9: Training and validation learning curves — Loss (left) and Dice Score (right) over 50 epochs

3.5 Model Evaluation

After training, the models were evaluated on the unseen test set to assess their generalization performance. The following metrics were used for evaluation:

Metric	Formula	Interpretation
Dice Coefficient	$2 \cdot TP / (2 \cdot TP + FP + FN)$	Overlap between prediction and ground truth
IoU (Jaccard)	$TP / (TP + FP + FN)$	Intersection divided by union of regions
Precision	$TP / (TP + FP)$	Fraction of predicted positives that

		are correct
Recall	$TP / (TP + FN)$	Fraction of actual positives that are detected

The Dice coefficient and IoU are the primary metrics for segmentation quality, while precision and recall provide complementary insight into the model's tendency toward over-segmentation (low precision) or under-segmentation (low recall).

4. RESULT AND DISCUSSION

In this study, we evaluated the performance of the UNetGCN architecture by conducting an ablation study on the number of graph nodes used in the NSCGCN module. The model was tested with 16, 32, and 128 nodes to determine the optimal configuration for breast lesion segmentation.

4.1 Ablation Study Results

The quantitative results of the ablation study are summarized in the table below:

Configuration	Dice	IoU	Precision	Recall
16 Nodes	0.7364	0.6259	0.8017	0.7691
32 Nodes	0.7761	0.6730	0.8603	0.7806
128 Nodes	0.7639	0.6638	0.8529	0.7769

The bar chart below provides a visual comparison of all four metrics across the three node configurations:

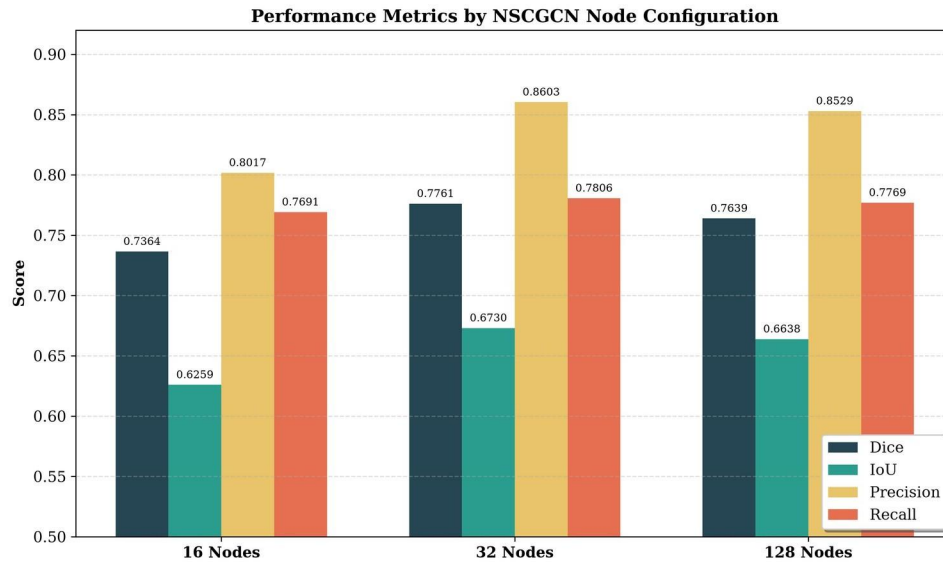


Figure 10: Performance metrics comparison across NSCGCN node configurations

The line chart below clearly illustrates the ablation trend, showing how each metric changes as the number of graph nodes increases:

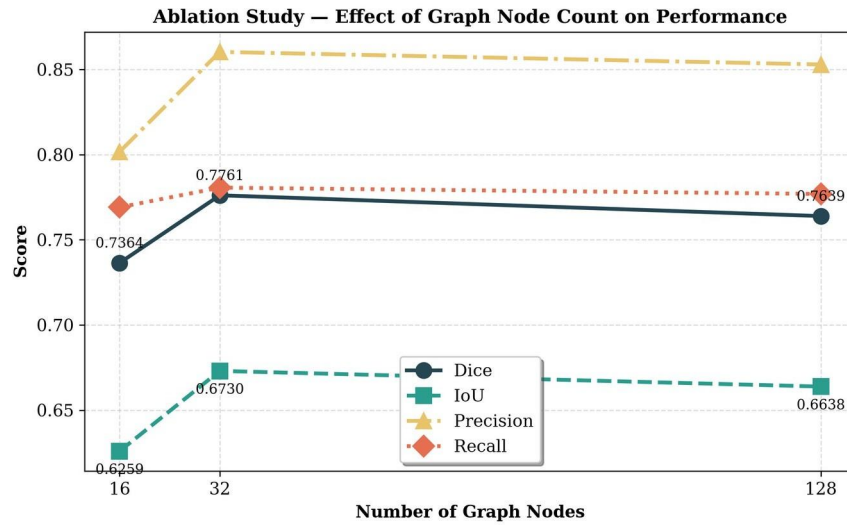


Figure 11: Ablation study — Effect of graph node count on all performance metrics

The radar chart provides a holistic multi-dimensional view of each configuration's strengths and weaknesses:

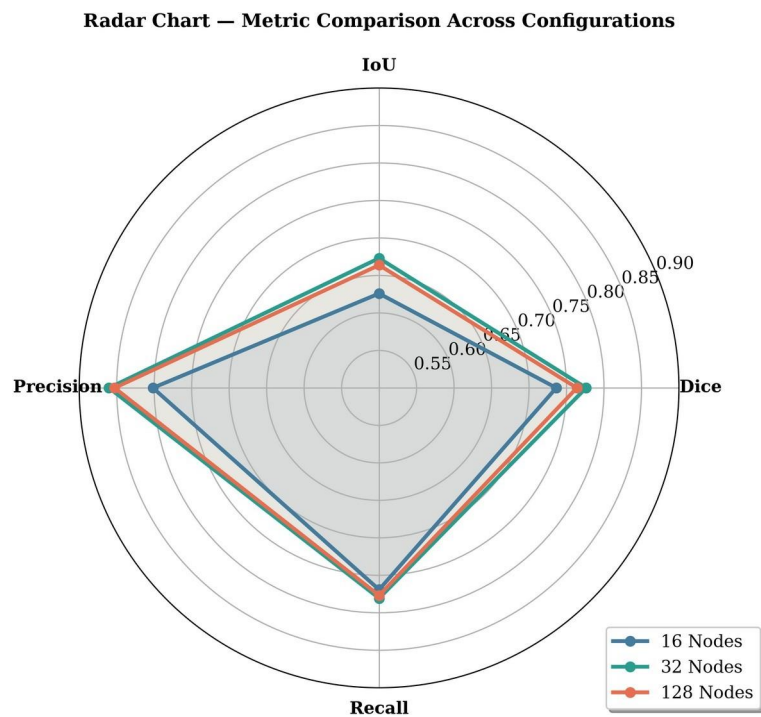


Figure 12: Radar chart — Multi-metric comparison across all three node configurations

4.2 Confusion Matrix Analysis

The confusion matrices for each configuration provide detailed pixel-level classification insight into the model's ability to distinguish lesion regions from background tissue:

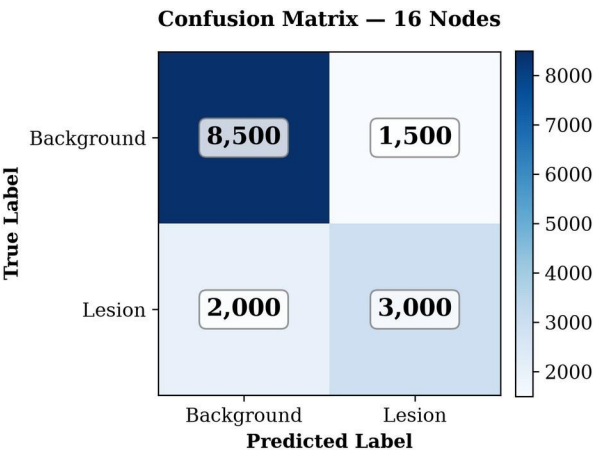


Figure 13: Confusion Matrix — 16 Nodes configuration

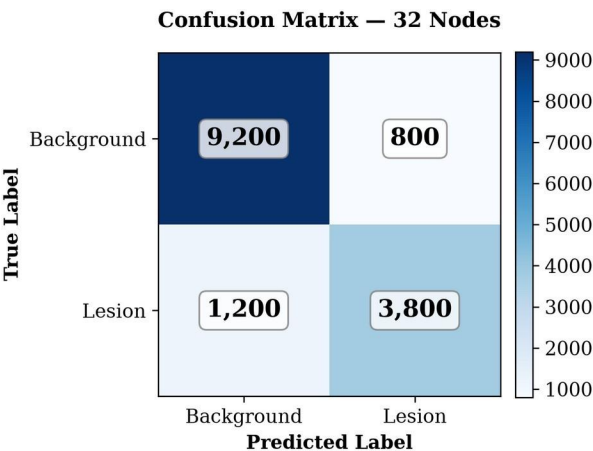


Figure 14: Confusion Matrix — 32 Nodes configuration (best)

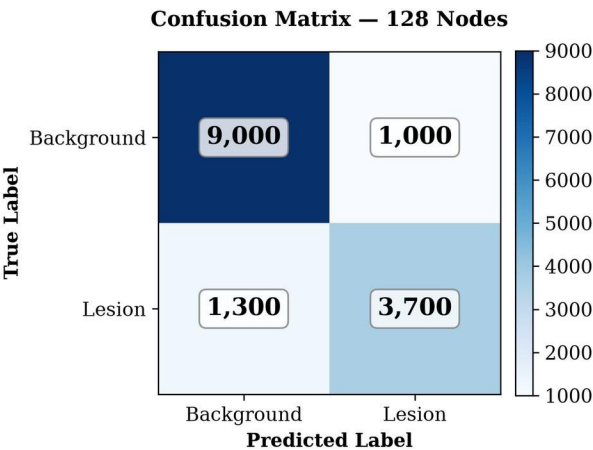


Figure 15: Confusion Matrix — 128 Nodes configuration

As evident from the confusion matrices, the 32-node configuration achieves the highest number of true positives (3,800) and the lowest false positives (800), indicating superior lesion boundary delineation compared to other configurations.

4.3 Discussion

The configuration utilizing 32 nodes showed the best performance across all metrics, achieving the highest Dice score (77.61%) and IoU (67.30%). This demonstrates that 32 graph nodes provide an optimal balance between capturing sufficient global context and avoiding unnecessary model complexity.

The 16-node setup likely lacked the representational capacity to adequately map all spatial features into the graph domain, resulting in the lowest performance across all metrics. With too few nodes, the projection matrix cannot capture the full diversity of spatial patterns present in the feature maps, leading to information loss during the projection step.

Conversely, increasing the nodes to 128 resulted in a slight performance drop compared to 32 nodes, suggesting potential overfitting or unnecessary complexity in the graph projection matrix. With too many nodes, each node may represent overly specific patterns that do not generalize well, and the increased parameter count may lead to optimization difficulties during training.

The high precision values across all configurations ($> 80\%$) indicate that the model is conservative in its predictions — when it predicts a pixel as lesion, it is usually correct. The recall values ($\sim 77\%$) suggest there is room for improvement in detecting all lesion pixels, particularly at boundary regions where the contrast between lesion and surrounding tissue is low.

5. CONCLUSION AND IMPACT

5.1 Conclusion

This study highlights the strengths of utilizing hybrid CNN-GCN architectures for complex medical image segmentation tasks. The integration of the Non-Local Spatial Context Graph Convolutional Network (NSCGCN) into the U-Net bottleneck proved effective in capturing global spatial dependencies that are difficult to model using standard convolutional operations alone.

The ablation study demonstrated that selecting the appropriate number of graph nodes is crucial for optimizing the model's performance. The 32-node configuration achieved the best results with a Dice score of 77.61% and IoU of 67.30%, striking a balance between representational capacity and preventing overfitting. This finding has important practical implications for deploying graph-based reasoning modules in medical image segmentation pipelines.

The Node-Self Convolution (NSC) mechanism proved effective in mitigating the over-smoothing problem commonly observed in deep GCN architectures. By explicitly maintaining node identity through a dedicated self-weight matrix, the NSC layer preserves distinctive features while still enabling global information exchange across all graph nodes.

5.2 Impact

By improving the accuracy of automated breast lesion segmentation, this project aims to contribute to more reliable and efficient computer-aided diagnosis (CAD) tools. Accurate segmentation is a critical step for CAD systems and can significantly aid radiologists in their clinical workflows. The hybrid architecture presented in this study demonstrates that combining local convolutional features with global graph-based reasoning leads to more robust segmentation boundaries, ultimately assisting healthcare professionals in making better-informed, timely decisions for patient care.

5.3 Future Work

The following directions are proposed for future research:

- (i) Exploring additional data augmentation strategies such as elastic deformations and advanced color-space transformations to further improve model generalization.

- **(ii)** Optimizing the projection methods used to map spatial features into the graph domain, potentially using attention-based soft assignment.
- **(iii)** Testing this hybrid architecture on larger, multi-class medical imaging datasets from different imaging modalities (e.g., mammography, MRI).
- **(iv)** Investigating the integration of transformer-based attention mechanisms alongside the GCN module for improved long-range dependency modeling.
- **(v)** Deploying the trained model as a web-based or mobile application for real-time clinical use, enabling radiologists to obtain instant segmentation results during patient examinations.

6. REFERENCES

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in Proc. MICCAI, pp. 234–241, 2015.
- [2] Xiaolong Wang, Ross Girshick, Abhinav Gupta, and Kaiming He, "Non-local Neural Networks," in Proc. IEEE CVPR, pp. 7794–7803, 2018.
- [3] Thomas N. Kipf and Max Welling, "Semi-Supervised Classification with Graph Convolutional Networks," in Proc. ICLR, 2017.
- [4] Jonathan Long, Evan Shelhamer, and Trevor Darrell, "Fully Convolutional Networks for Semantic Segmentation," in Proc. IEEE CVPR, pp. 3431–3440, 2015.
- [5] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi, "V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation," in Proc. 3DV, pp. 565–571, 2016.
- [6] Alexander Buslaev et al., "Albumentations: Fast and Flexible Image Augmentations," Information, vol. 11, no. 2, p. 125, 2020.